Subject: **CAPE Pure Mathematics Unit 1**

Topic: **Past Paper June 2024 – Paper 3 – Solutions**

June 2024 – Question 1 – Solution

(a) The number of cases of a particular disease can be modelled by the equation $A = A_0e^{kt}$ where A is the number of cases at time t (in years) and A_0 is the number of cases when $t = 0$. It is known that there are 10 000 cases of the disease at present and that the number of cases decrease by 10% each year. How many years must elapse for there to be 500 cases of the disease? [9]

$A = A_0e^{kt}$

At $t = 1$,

$A = A_0e^{k(1)}$

$A = A_0e^{k}$ → Equation 1

We are given that the number of cases decrease by 10% each year.

When $t = 1$,

$A = 90\%$ of A_0

$A = 0.9 \times A_0$

$A = 0.9A_0$ → Equation 2

Equating both equations gives:

$A_0e^{k} = 0.9A_0$

$e^{k} = 0.9$

Taking ln on both sides gives:

$\ln e^k = \ln 0.9$

$k \ln e = \ln 0.9$

$k(1) = \ln 0.9$

$k = \ln 0.9$

So, we have,

$A = A_0 e^{(\ln 0.9)t}$

When $t = 0, A_0 = 10\ 000$.

The number of cases of a particular disease can be modelled by the equation:

$A = 10\ 000 e^{(\ln 0.9)t}$

When $A = 500$,

$500 = 10\ 000 e^{(\ln 0.9)t}$

$\frac{500}{10\ 000} = e^{(\ln 0.9)t}$

$\frac{1}{20} = \left(e^{\ln 0.9}\right)^t$

$\frac{1}{20} = (0.9)^t$

$(0.9)^t = \frac{1}{20}$

Taking ln on both sides gives:

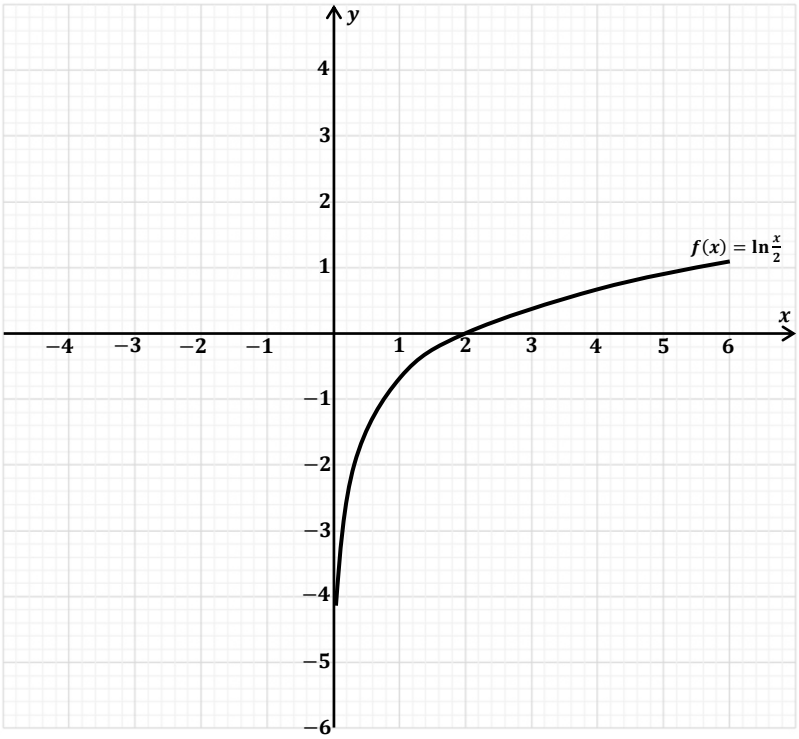
$\ln(0.9)^t = \ln\left(\frac{1}{20}\right)$

$t \ln 0.9 = \ln\left(\frac{1}{20}\right)$

$t = \frac{\ln\left(\frac{1}{20}\right)}{\ln 0.9}$

$t = 29$ years (to the nearest whole number)

(b) The diagram below shows the graph of the function $f(x) = \ln\frac{x}{2}$.



On the same axes,

(i) sketch and label the graph of f^{-1} [4]

$f(x) = \ln\frac{x}{2}$

Let $y = f(x)$.

$y = \ln\frac{x}{2}$

Interchange the variables.

$x = \ln\frac{y}{2}$

Make y the subject of the formula.

$$x = \ln\frac{y}{2}$$

$$e^x = e^{\ln\frac{y}{2}}$$

$$e^x = \frac{y}{2}$$

$$2e^x = y$$

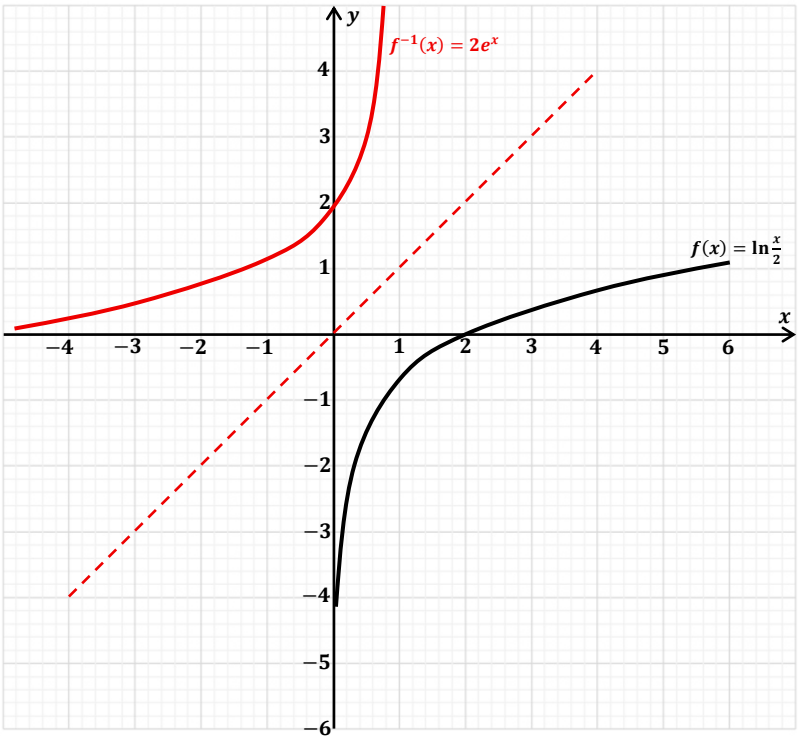
So, $f^{-1}(x) = 2e^x$.

The graph of $f^{-1}(x)$ is the reflection of f in the line $y = x$.

From the diagram above, the graph of $f(x)$ passes through the point $(2, 0)$.

So, the graph of $f^{-1}(x)$ will pass through the point $(0, 2)$.

The sketch is shown below:



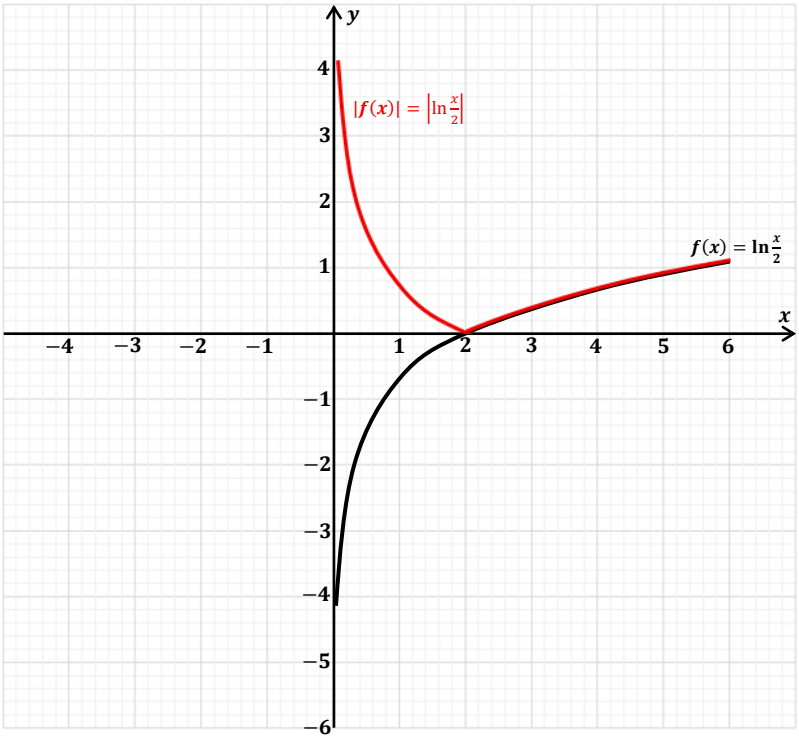
(ii) sketch and label the graph of $|f|$ [3]

$f(x) = \ln\frac{x}{2}$

$|f(x)| = \left|\ln\frac{x}{2}\right|$

To sketch the graph of $|f|$, we reflect the part below the x-axis, that is, $x < 2$, above the x-axis.

The sketch is shown below:



(c) Given that $\log 2 = 0.3010$ and $\log 3 = 0.4771$, calculate the value of $\log 216$.

Show all working and provide your answer using 3 significant figures. [4]

$$216 = 36 \times 6$$
$$= 6 \times 6 \times 6$$
$$= 6^3$$
$$= (2 \times 3)^3$$
$$= 2^3 \cdot 3^3$$

Now,

$$\log 216 = \log(2^3 \cdot 3^3)$$
$$= \log 2^3 + \log 3^3$$
$$= 3\log 2 + 3\log 3$$
$$= 3(0.3010) + 3(0.4771)$$
$$= 0.903 + 1.4313$$
$$= 2.3343$$
$$= 2.33 \quad \text{(to 3 significant figures)}$$

$\therefore$ The value of $\log 216 = 2.33$ (to 3 significant figures).

Total: 20 marks

June 2024 – Question 2 – Solution

(a) The route for a drone flying competition is mapped onto the coordinate axes in such a way that the finish line runs through the points $A(11, 3, -1)$ and $B(-9, 2, 4)$. One of the drone pilots decides to fly his drone in a straight path perpendicular to the finish line. Assume that he begins flying his drone from the point $(0, 0, 0)$.

What are the coordinates of the point, P, at which his drone will cross the finish line? [8]

Points are $A(11, 3, -1)$ and $B(-9, 2, 4)$.

The position vectors are:

$$\overrightarrow{OA} = \begin{pmatrix} 11 \\ 3 \\ -1 \end{pmatrix} \quad \text{and} \quad \overrightarrow{OB} = \begin{pmatrix} -9 \\ 2 \\ 4 \end{pmatrix}$$

Using the triangle law,

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$$

$$= \begin{pmatrix} -9 \\ 2 \\ 4 \end{pmatrix} - \begin{pmatrix} 11 \\ 3 \\ -1 \end{pmatrix}$$

$$= \begin{pmatrix} -20 \\ -1 \\ 5 \end{pmatrix}$$

Let $\boldsymbol{a} = \begin{pmatrix} 11 \\ 3 \\ -1 \end{pmatrix}$ and $\boldsymbol{n} = \begin{pmatrix} -20 \\ -1 \\ 5 \end{pmatrix}$.

The vector equation of the line is:

$$\boldsymbol{r} = \boldsymbol{a} + \lambda \boldsymbol{n}$$

$$\boldsymbol{r} = \begin{pmatrix} 11 \\ 3 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} -20 \\ -1 \\ 5 \end{pmatrix}$$

$$\boldsymbol{r} = \begin{pmatrix} 11 \\ 3 \\ -1 \end{pmatrix} + \begin{pmatrix} -20\lambda \\ -\lambda \\ 5\lambda \end{pmatrix}$$

$$\boldsymbol{r} = \begin{pmatrix} 11 - 20\lambda \\ 3 - \lambda \\ -1 + 5\lambda \end{pmatrix}$$

The drone's path from the origin to P is perpendicular to the finish line.

So, we have,

$$\boldsymbol{r} \cdot \boldsymbol{n} = 0$$

$$\begin{pmatrix} 11 - 20\lambda \\ 3 - \lambda \\ -1 + 5\lambda \end{pmatrix} \cdot \begin{pmatrix} -20 \\ -1 \\ 5 \end{pmatrix} = 0$$

$$(11 - 20\lambda)(-20) + (3 - \lambda)(-1) + (-1 + 5\lambda)(5) = 0$$

$$-220 + 400\lambda - 3 + \lambda + -5 + 25\lambda = 0$$

$$426\lambda - 228 = 0$$

$$426\lambda = 228$$

$$\lambda = \frac{228}{426}$$

$$\lambda = \frac{38}{71}$$

Let the coordinates of P be (x, y, z).

The position vector is $\overrightarrow{OP} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$.

Now,

$$\boldsymbol{r} = \begin{pmatrix} 11 - 20\lambda \\ 3 - \lambda \\ -1 + 5\lambda \end{pmatrix}$$

The Parametric equations are:

$x = 11 - 20\lambda$

$y = 3 - \lambda$

$z = -1 + 5\lambda$

When $\lambda = \frac{38}{71}$,

$x = 11 - 20\left(\frac{38}{71}\right)$

$= 11 - \frac{760}{71}$

$= \frac{21}{71}$

$y = 3 - \left(\frac{38}{71}\right)$

$= \frac{175}{71}$

$z = -1 + 5\left(\frac{38}{71}\right)$

$= -1 + \frac{190}{71}$

$= \frac{119}{71}$

$\therefore$ The coordinates of the point, P, at which his drone will cross the finish line is:

$P\left(\frac{21}{71}, \frac{175}{71}, \frac{119}{71}\right)$

(b) Let $f(x) = \sin x - 3\cos x$.

(i) Express f in the form $R\cos(x+\alpha)$, where $R > 0$. [7]

Let $\sin x - 3\cos x = R\cos(x+\alpha)$

$-3\cos x + \sin x = R\cos x\cos\alpha - R\sin x\sin\alpha$

Equating coefficients gives:

$R\cos\alpha = -3$ → Equation 1

$R\sin\alpha = -1$ → Equation 2

Equation 2 ÷ Equation 1 gives:

$\frac{R\sin\alpha}{R\cos\alpha} = \frac{-1}{-3}$

$\tan\alpha = \frac{1}{3}$

$\alpha = \tan^{-1}\left(\frac{1}{3}\right)$

$\alpha = 0.322^c$

$R = \sqrt{(-3)^2 + (1)^2}$

$= \sqrt{9+1}$

$= \sqrt{10}$

$\therefore \sin x - 3\cos x = \sqrt{10}\cos(\theta + 0.322^c)$

(ii) Hence, or otherwise, obtain the general solution of $\sin x - 3\cos x = 2$. [5]

$$\sin x - 3\cos x = 2$$

$$\sqrt{10}\cos(\theta + 0.322^c) = 2$$

$$\cos(\theta + 0.322^c) = \frac{2}{\sqrt{10}}$$

$$PV = 0.866^c$$

$\therefore$ G.S. is $\theta + 0.322^c = 2n\pi \pm 0.866^c,\ n \in \mathbb{Z}$

$$\theta = 2n\pi \pm 0.866^c - 0.322^c,\ n \in \mathbb{Z}$$

Total: 20 marks

June 2024 – Question 3 – Solution

(a) A spherical weather balloon with a radius, r, of 2 metres is being inflated by an electric pump.

(i) Calculate the rate at which the balloon is being inflated if its radius is increasing at a rate of 1 centimetre per second.

Note that the volume of a sphere is $V = \frac{4}{3}\pi r^3$. [4]

We are given that:

$$\frac{dr}{dt} = 1\ cms^{-1}$$

$$= \frac{1}{100}\ ms^{-1}$$

$$= 0.01\ ms^{-1}$$

Now,

$$V = \frac{4}{3}\pi r^3$$

$$\frac{dV}{dr} = \frac{4}{3}(3)\pi r^{3-1}$$

$$= 4\pi r^2$$

When $r = 2$,

$$\frac{dV}{dr} = 4\pi(2)^2$$

$$= 4\pi \times 4$$

$$= 16\pi$$

Using chain rule,

$$\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$$

$$= 16\pi \times 0.01$$

$$= 0.16\pi$$

$\therefore$ The rate at which the balloon is being inflated is $0.16\pi\ cm^3 s^{-1}$.

(ii) If air continues to be pumped into the balloon at the same rate, what will the rate of change of its radius be when the radius is 6 metres? [3]

$$\frac{dV}{dt} = 0.16\pi$$

$$\frac{dV}{dr} = 4\pi r^2$$

When $r = 6$,

$$\frac{dV}{dr} = 4\pi(6)^2$$

$$= 4\pi \times 36$$

$$= 144\pi$$

Using chain rule,

$$\frac{dr}{dt} = \frac{dr}{dV} \times \frac{dV}{dt}$$

$$= \frac{1}{900}$$

$\therefore$ The rate of change of its radius will be $\frac{1}{900}\ ms^{-1}$.

(b) When the path of a roller coaster at a theme park is superimposed onto a graph, it has a gradient that is proportional to the product of the x and y coordinates. The path is known to pass through the points $(4, 1)$ and $(-2, e^2)$.

Determine the equation of the curve of the roller coaster.

You may use the fact that $\int \frac{1}{x} dx = \ln x + c$. [8]

The gradient is proportional to the product of the x and y coordinates.

$$\frac{dy}{dx} \propto xy$$

$$\frac{dy}{dx} = kxy$$

$$\frac{dy}{dx} = \frac{kx}{\left(\frac{1}{y}\right)}$$

Separating variables:

$$\frac{1}{y} dy = kx\, dx$$

Integrating both sides gives:

$$\int \frac{1}{y} dy = \int kx\, dx$$

$$\ln y = \frac{kx^2}{2} + c$$

$$e^{\ln y} = e^{\frac{kx^2}{2} + c}$$

$$y = e^{\frac{kx^2}{2} + c}$$

Substituting the point $(4, 1)$ gives:

$$1 = e^{\frac{k(4)^2}{2} + c}$$

$$1 = e^{8k+c} \qquad \rightarrow \text{Equation 1}$$

Substituting the point $(-2, e^2)$ gives:

$$e^2 = e^{\frac{k(-2)^2}{2} + c}$$

$$e^2 = e^{2k+c} \qquad \rightarrow \text{Equation 2}$$

Rearranging Equation 2 gives:

$$e^2 = e^{2k} \cdot e^c$$

$$\frac{e^2}{e^c} = e^{2k} \qquad \rightarrow \text{Equation 3}$$

Rearranging Equation 1 gives:

$$1 = e^{8k} \cdot e^c$$

$$\frac{1}{e^c} = e^{8k}$$

$$\frac{1}{e^c} = (e^{2k})^4 \qquad \rightarrow \text{Equation 4}$$

Substituting Equation 3 into Equation 4 gives:

$$\frac{1}{e^c} = \left(\frac{e^2}{e^c}\right)^4$$

$$\frac{1}{e^c} = \frac{e^8}{e^{4c}}$$

$$e^{4c} = e^8 \times e^c$$

$$\frac{e^{4c}}{e^c} = e^8$$

$$e^{3c} = e^8$$

$$(e^c)^3 = e^8$$

$$e^c = (e^8)^{\frac{1}{3}}$$

$$e^c = e^{\frac{8}{3}}$$

Since the bases are the same, we can equate the powers.

$c = \frac{8}{3}$

Substituting $e^c = e^{\frac{8}{3}}$ into Equation 3 gives:

$$\frac{e^2}{e^{\frac{8}{3}}} = e^{2k}$$

$$e^{2-\frac{8}{3}} = e^{2k}$$

$$e^{\frac{6}{3}-\frac{8}{3}} = e^{2k}$$

$$e^{-\frac{2}{3}} = e^{2k}$$

Since the bases are the same, we can equate the powers.

$$-\frac{2}{3} = 2k$$

$$-\frac{2}{3} \div 2 = k$$

$$-\frac{2}{3} \times \frac{1}{2} = k$$

$$-\frac{1}{3} = k$$

We have $y = e^{\frac{kx^2}{2} + c}$.

Substituting $k = -\frac{1}{3}$ and $c = \frac{8}{3}$ into $y = e^{\frac{kx^2}{2} + c}$ gives:

$$y = e^{\frac{\left(-\frac{1}{3}\right)x^2}{2} + \frac{8}{3}}$$

$$= e^{-\frac{1}{6}x^2 + \frac{8}{3}}$$

$$= e^{\frac{8}{3} - \frac{x^2}{6}}$$

$\therefore$ The equation of the curve of the roller coaster is: $y = e^{\frac{8}{3} - \frac{x^2}{6}}$

(c) Calculate the area enclosed between the parabola $f(x) = x^2 - 3x - 4$ and the line $g(x) = x + 1$. [5]

$y = x^2 - 3x - 4$ → Equation 1

$y = x + 1$ → Equation 2

Equating both equations gives:

$$x^2 - 3x - 4 = x + 1$$

$$x^2 - 3x - x - 4 - 1 = 0$$

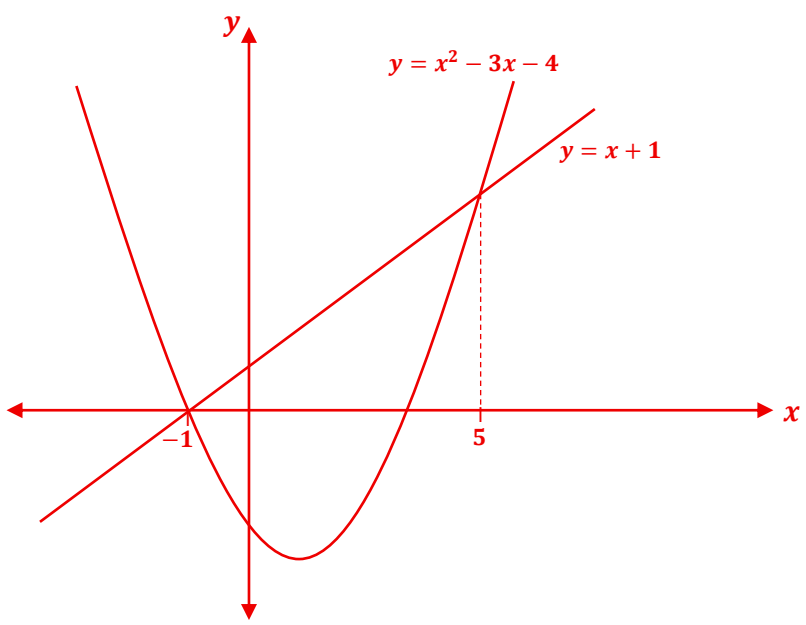
$$x^2 - 4x - 5 = 0$$

$$(x + 1)(x - 5) = 0$$

Either $x + 1 = 0$ or $x - 5 = 0$

$x = -1$ $x = 5$

Consider the sketch below:



Enclosed area = Area under the line − Area under the curve

$$= \int_{-1}^{5}(x+1)\,dx - \int_{-1}^{5}(x^2-3x-4)\,dx$$

$$= \int_{-1}^{5}[(x+1)-(x^2-3x-4)]\,dx$$

$$= \int_{-1}^{5}(x+1-x^2+3x+4)\,dx$$

$$= \int_{-1}^{5}(-x^2+4x+5)\,dx$$

$$= \left[-\frac{x^3}{3}+2x^2+5x\right]_{-1}^{5}$$

$$= \left[-\frac{(5)^3}{3}+2(5)^2+5(5)\right]-\left[-\frac{(-1)^3}{3}+2(-1)^2+5(-1)\right]$$

$$= \left(-\frac{125}{3}+50+25\right)-\left(\frac{1}{3}+2-5\right)$$

$$= \frac{100}{3}-\left(-\frac{8}{3}\right)$$

$$= \frac{100}{3}+\frac{8}{3}$$

$$= \frac{108}{3}$$

$$= 36 \text{ units}^2$$

∴ The area enclosed between the parabola and the line is 36 square units.

Total: 20 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.